
Permanent Address

Koufokosta 2, 35131
Lamia, GRC
(0030) 697-320 7886

CURRENT POSITION

University of Thessaly

Post Doctoral Researcher in Computer Science and Biomedical Informatics

– Advisor: Tasoulis Sotiris

Lamia, GRC

February 2024–Current

EDUCATION

University of Thessaly

Doctor of Philosophy (Ph.D.) in Computer Science and Biomedical Informatics

– Advisor: Tasoulis Sotiris

– Thesis: “Design and Implementation of Machine Learning Algorithms in Big Biomedical Data”

Lamia, GRC

2019–2023

University of Thessaly

Certificate of Pedagogical & Teaching Competence (Παιδαγωγικής & Διδακτικής Επάρκειας)

Lamia, GRC

2022–2023

University of Patras

Bachelor with Integrated master in Computer Engineering and Informatics, Degree: 6.70

– Thesis: “Knowledge mining and visualization of molecular biology networks”

Patras, GRC

2009–2019

EXPERIENCE

University of Thessaly

Post Doctoral Researcher in Computer Science and Biomedical Informatics

– Advisor: Tasoulis Sotiris

– Title: “Machine Learning Algorithms in Big Data”

Lamia, GRC

February 2024–Current

University of Thessaly

Role: Researcher

– “Bridging big omic, genetic and medical data for Precision Medicine implementation in Greece (project code TAEDR-0539180)”

– Funded by the European Union – NextGenerationEU through Greece 2.0—National Recovery and Resilience Plan, under the call “Flagship actions in interdisciplinary scientific fields with a special focus on the productive fabric” (ID 16618).

Lamia, GRC

June 2024–Current

Institute for Bio-economy and Agri-technology iBO/CERTH

Role: Researcher

– “Personalized Rehabilitation Via Novel AI Patient Stratification Strategies (PREPARE)”

– Funded by the European Union. UK participants in Horizon Europe Project PREPARE are supported by UKRI grant number 10086219 (Trilateral Research).

Lamia, GRC

December 2023–Current

Institute for Bio-economy and Agri-technology iBO/CERTH

Role: Researcher

Lamia, GRC

August 2021–November 2023

- “uPrevent”
- Co-funded by the European Union and the European Regional Development Fund under the Single Action for State Aid for Research, Technological Development & Innovation “RESEARCH–CREATE–INNOVATE”.

Signal Ocean SMPC

Lamia, GRC

Role: Researcher

February 2023–October 2023

- “Design and implementation of machine learning algorithms for the prediction and analysis of the course of transport vessels”
- Funded by the company Signal Ocean SMPC.

University of Thessaly

Lamia, GRC

Role: Researcher

March 2022–September 2023

- “ParICT_CENG: Improving ICT research infrastructures in Central Greece for processing large volumes of data from sensor streams, multimedia and complex mathematical simulation models”
- Funded under the NSRF 2014-2020, co-financed by Greece and the European Union (European Regional Development Fund).

Signal Ocean SMPC

Lamia, GRC

Role: Researcher

March 2021–December 2021

- “Design and implementation of machine learning algorithms for the prediction and analysis of the course of transport vessels”
- Funded by the company Signal Ocean SMPC.

Institute for Bio-economy and Agri-technology iBO/CERTH

Lamia, GRC

Role: Researcher

March 2016–April 2021

- “Advanced personalised, multi-scale computer models preventing OsteoArthritis,”
- Funded by European Community’s H2020 Programme, under grant agreement Nr. 777159.

PUBLICATIONS

- [1] P. Barmpas, **P. Anagnostou**, S. K. Tasoulis, V. P. Plagianakos, and S. V. Georgakopoulos, “Efficient hybrid hierarchical clustering with incremental silhouette score for large, noisy datasets”, *International Journal of Neural Systems*, 2025.
- [2] A. Biloni, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “Enhancing machine learning models for medical coding: Diagnostic codes mapping and synthetic clinical notes”, in *2025 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, IEEE, 2025, pp. 1–8.
- [3] P. C. Theocharopoulos, **P. Anagnostou**, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Large language models for efficient topic modeling”, *Neural Computing and Applications*, pp. 1–19, 2025.
- [4] K. Tsiaktani, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, J. A. Gittings, D. E. Raitsos, A. Sergounioti, and V. P. Plagianakos, “Short-term impacts of weather conditions on biomarkers among urban adults in greece”, in *2025 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, 2025, pp. 1–8.
- [5] **P. Anagnostou**, S. K. Tasoulis, A. G. Vrahatis, S. V. Georgakopoulos, and V. P. Plagianakos, “Boosting neural network performance for high dimensional data through random projections”, *Available at SSRN 4991467*, 2024.

- [6] P. Barmpas, **P. Anagnostou**, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “HCER: Hierarchical clustering-ensemble regressor”, in *Engineering Applications of Neural Networks*, L. Iliadis, I. Maglogiannis, A. Papaleonidas, E. Pimenidis, and C. Jayne, Eds., Cham: Springer Nature Switzerland, 2024, pp. 369–378, ISBN: 978-3-031-62495-7.
- [7] **P. Anagnostou**, P. Barmpas, S. K. Tasoulis, S. V. Georgakopoulos, and V. P. Plagianakos, “Neural networks voting for projection based ensemble classifiers”, in *2023 IEEE International Conference on Big Data (BigData)*, IEEE, 2023, pp. 4567–4574.
- [8] **P. Anagnostou**, N. G. Pavlidis, and S. K. Tasoulis, “Ensemble clustering for boundary detection in high-dimensional data”, in *Proceedings of the 9th Annual Conference on Machine Learning, Optimization and Data Science (LOD)*, Lake District, UK: Springer, 2023.
- [9] **P. Anagnostou**, S. Tasoulis, V. P. Plagianakos, and D. Tasoulis, “HiPart: Hierarchical divisive clustering toolbox”, *Journal of Open Source Software*, vol. 8, no. 84, p. 5024, 2023.
- [10] P. C. Theocharopoulos, **P. Anagnostou**, A. Tsoukala, S. V. Georgakopoulos, S. K. Tasoulis, and V. P. Plagianakos, “Detection of fake generated scientific abstracts”, in *2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService)*, 2023, pp. 33–39.
- [11] P. Barmpas, S. Tasoulis, A. G. Vrahatis, S. V. Georgakopoulos, **P. Anagnostou**, M. Prina, J. L. Ayuso-Mateos, J. Bickenbach, I. Bayes, M. Bobak, *et al.*, “A divisive hierarchical clustering methodology for enhancing the ensemble prediction power in large scale population studies: The athlos project”, *Health Information Science and Systems*, vol. 10, no. 1, p. 6, 2022.
- [12] **P. Anagnostou**, S. Tasoulis, A. G. Vrahatis, S. Georgakopoulos, M. Prina, J. L. Ayuso-Mateos, J. Bickenbach, I. Bayes-Marin, F. F. Caballero, L. Egea-Cortés, *et al.*, “Enhancing the human health status prediction: The athlos project”, *Applied Artificial Intelligence*, vol. 35, no. 11, pp. 834–856, 2021.
- [13] **P. Anagnostou**, P. Barbas, A. G. Vrahatis, and S. K. Tasoulis, “Approximate knn classification for biomedical data”, in *2020 IEEE International Conference on Big Data (Big Data)*, IEEE, 2020, pp. 3602–3607.

INVITED TALKS

- N. G. Pavlidis, **P. Anagnostou**, and S. Tasoulis, “Dimensionality reduction and clustering”, in Proceedings 35th Panhellenic & 1st International Statistics Conference, Athens, Greece, 2023. Co-presenters: Pavlidis, Nicos G., and Tasoulis, Sotiris.

REFeree SERVICE

- Reviewer on, the journal of Pattern Recognition Letters (PRLetters) the journal of Future generation computer systems (FGCS), the Journal of Open-Source Software (JOSS), the journal of Neural computing & applications (NCAA), the IEEE Congress on Evolutionary Computation (IEEE CEC), and the IEEE Symposium Series on Computational Intelligence (IEEE SSCI).
- **Program Committee Member**, IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (IEEE CIBCB), 2025
- **Program Committee Member**, IEEE International Conference on Big Data (IEEE BigData), 2021 and 2023

TEACHING

- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2025
Linked Computing and Data Systems (DIB_U_132)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.

- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2025
Ubiquitous Computing Applications Programming (DIB_U_235)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Fall 2024
Microprocessors (DIB_U_206)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Fall 2024
Internet Application Technologies (DIB_U_182)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2024
Embedded Computer Systems in Biomedical Engineering (DIB_U_211)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.
- **Instructor** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2024
Ubiquitous Computing Applications Programming (DIB_U_235)
- Part of the program “Acquisition of Academic Teaching Experience for New PhD Holders” co-funded by the European Union and National Resources.
- **Teaching Assistant** at University of Thessaly’s School of Science Fall 2019 - 2024
Educational software principals (DIB_P_112)
- **Teaching Assistant** at Department of Computer Science and Biomedical Informatics, University of Thessaly Spring 2021 - 2023
Deep Learning (DIB_U_240)
- **Teaching Assistant** at University of Thessaly’s School of Science Spring 2020 - 2022
Computer programming subjects (DIB_P_130)
- **Laboratory assistant** at Department of Computer Engineering and Informatics, University of Patras Spring 2017
Digital Electronics Laboratory (CEID_NY166)

TECHNICAL SKILLS

- **Programming Languages:**
 - **Object Oriented:** C++, Java
 - **Procedural:** C, Pascal
 - **Scripting:** R, Python, Bash, PHP
 - **Other:** Latex, Matlab, XML, HTML5, CSS, MySQL, Prolog
- **Operating Systems:**
 - GNU/Linux. Proficient in using CLI and GUI.
 - Microsoft Windows
 - Android. Application development.
- **Version Control System:** Git
- **Build automation tool:** Apache Maven

LANGUAGES

- **Greek:** Native Language
- **English:** State Certificate of Language Proficiency (KPG), Level B2